Sensitivity Comparison of Atomic Interferometers for Gravitational Measurements on Positronium Using Monte Carlo Simulations

or Treatise on Shelling Bananas

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contents

problem

figure: a banana

what is a positronium?

$$\underbrace{\text{positr}}_{e^+ \text{ and } e^-}\;\underbrace{\text{onium}}_{\text{bound state}}$$

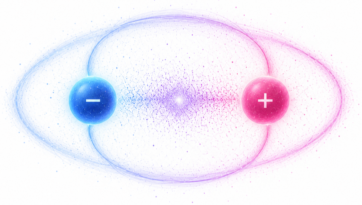


figure: artistic image of positronium

positronium states

and a quick repeat from atomic physics

- para-positronium 1^1S_0
- ortho-positronium 1^3S_1
- meta-stable-positronium 2^3S_1

what does this mean?

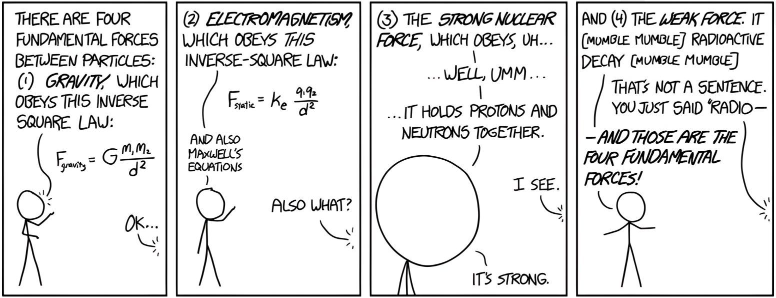
$$n^{2S+1}L_{m_S}$$

$$\vec{J} = \vec{L} + \vec{S}$$

how to obtain meta-stable state?

it's hard

fundamental forces in physics



fundamental forces in physics

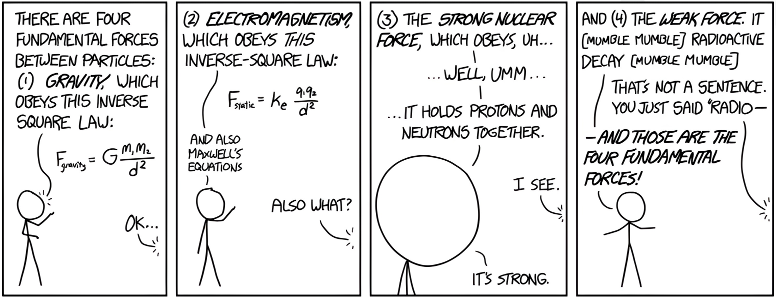


figure: "Of these four forces, there's one we don't really understand." "Is it the weak force or the strong–" "It's gravity."

why do we bother?

> *'But in any event, it is in the hands of our generation to perform an experiment to measure the gravitational acceleration of antimatter. Some day it will be done, whether we do it or not. It will be done. If we do not do it, and the answer eventually turns out to be what we expect, then future generations will look back upon us and say it was a shame. But if the answer turns out to be a surprise, then, if we do not do it, future generations will look back upon us and say we were fools.'*

– M. M. Nieto et al. Theoretical Motivation for Gravitation Experiments on UltraLow Energy Antiprotons and Antihydrogen. arXiv, 1994.

a bit of theory

why are we even bothering to measure it?

- **CPT symmetry:** according to the CPT theorem, antimatter should experience the same gravitational acceleration as matter
- **photon behaviour:** photons are their own antiparticles, and they act just like General Relativity laws tell them so
- **conservation of energy:** if antimatter goes up, we can annihilate it "higher" than it was produced

a bit of theory...

However...

- **Kowitt theory:** antiparticles are "holes" in Dirac particle sea with negative gravitational mass
- **Santilli-Villata theory:** antimatter exists in inverted space-time, which is projected to our reality as negative motion
- **Cabbolet's theory:** particles move in "stepwise" motion, alternating between state of rest and "wave–like" movement
- **dark matter???**

Moiré deflectometer

Mach-Zehnder interferometer

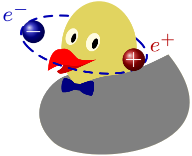
Talbot-Lau interferometer

Stern-Gerlach interferometer

why is it hard

bibliography

the end



(backup) why is it hard 2: electric boogaloo

(or why is it hard for me)